Haute École Arc Ingénierie
Filière : Informatique et systèmes de communication
Orientation: Ingénierie des données

Appendix

Finding the hydrogen 21-cm signal

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Présenté en vue de l'obtention du Bachelor of Science HES-SO



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Appendix A

Mandat

A.1 Titre

Exploring the Universe's Formation: Infer the Reionization Properties of the Hydrogen-21cm Signal

A.2 Contexte

A.2.1 SKAO and its goal

The Square Kilometer Array Observatory (SKAO) is an ambitious project which aims at creating an array of observatories in a one square kilometer in Australia and South Africa, which should be completed in 2030. The SKA project is an unprecedented ambitious project and global effort in radio astronomy to delve deeper and faster into the mysteries of the universe and could provide valuable insights into key periods of the universe creation, such as the cosmic Dark Ages, the cosmic dawn, and the epoch of reionisation. Funded in 2019, the SKAO Consortium has been created to support the preparation and processing of SKAO data. From 2022, Switzerland is an active member of the SKAO Consortium, with various schools participating in its realization.

A.2.2 SKAO SDC series

In order to be able to process the expected large and complex amounts of data that the SKAO will produce, a series of challenges, called SKAO Science Data Challenges (SDC) were created. These challenges consider difficult parts of data processing which need state-of-the-art methods to be addressed properly. These challenges are designed to be Data-driven, meaning even without any astrophysical knowledge you are able to participate and produce results.

A.3 Objectif

SKA 3rd SDC and the Hydrogen 21cm signal:

This project aims at addressing the third SDC challenge which aims at inferring the reionization properties of the Universe from power spectra of the hydrogen-21cm signal from the Epoch of Reionisation corresponding to different redshift ranges. This challenge will be addressed using mock datasets created by the SKAO Consortium to represent real data that the SKAO will produce. From an ML perspective, this is a regression task based on data from spectrogram.

A.4 Cahier des charges

- Discover the astrophysical domain enough to be able to produce AI models for this project
- Review the state of the art existing AI architecture and models to find the best suitable ones for this project

- Implement the best ones and fine-tune them
- Compare your score with the existing leaderboard
- Document and explain your work
- Depending on your interest: writing a short scientific paper

A.5 Ce que ce projet vous apportera

- Discover the world of astrophysics
- Apply AI to fundamental research
- Develop your academic skills
- Potentially discuss and present your results to an international consortium of researchers (if you want)

A.6 Questions/détails supplémentaires

N'hésitez pas à me contacter en cas de questions/clarifications nécessaires sur le sujet : emmanuel.desalis@he-arc.ch

État
Disponible

Proposé par
De Salis Emmanuel

Appendix B

Cahier des charges - 206 - Finding the H21cm signal

B.1 Situation initiale

Le consortium de SKAO a mis en marche des "data challenges" avec des données réalistes mais artificielles, pour but de "kickstarter" la préparation et le traitement de données lorsque l'observatoire est opérationnelle en 2030.

Le challenge de SDC3 étant fini, les équipes qui ont participé ont pu situer la précision de leur modèles d'inférer les propriétés de réionisation depuis les spectres de puissance (power spectra) des signaux de l'hydrogène 21-cm.

B.2 Buts visés

B.2.1 Buts principales

1. Phase litérature
    - Compréhension de la nature et objectif de SKAO et du data challenge SDC3.
    - Compréhension du dataset de PS1 et PS2.
    - Comprendre au moins un minimum de notion d'astrophysique pour mieux comprendre les datasets
2. Phase états de l'art
    - Comprendre les modèles utilisés par les équipes ayant eu un bon score pour les datasets de PS1 et PS2.
    - Comprendre des autres modèles.
3. Phase implémentation
    - Développer un modèle qui puisse inférer les propriétés de réionisation à partir des spectres de puissance des signaux de l'hydrogène 21-cm des datasets de PS1 et PS2.
    - Situer la précision (accuracy) des résultats en se basant sur le scoring du data challenge. Nous devons aussi tenir compte du fait que la précision sur les données artificielles ne garantit pas la fiabilité sur les données réelles.
4. Phase délivrables
    - Ecrire un rapport.
    - Préparer la présentation pour la défense.

B.2.2 Buts secondaires

- Utilisation de MLflow pour itérer, tester et évaluer plus rapidement les modèles.
- Implémenter un pipeline reproductible.

B.2.3 Buts nice-to-have

- Si nous estimons que le travail sur le challenge 3b est suffisant, il serait possible d'entamer le travail sur le challenge 3a, sur le traitement d'un dataset contaminé pour en extraire les spectres de puissance des signaux de l'hydrogène 21-cm.
- Au cas où les résultats sont soit encourageants, soit très mauvais, il serait possible d'écrire un papier scientifique et de le faire publier.
- Pouvoir faire évaluer un FITS cube par SKAO.

B.2.4 Contraintes éventuelles

- Le temps.
- Compréhension des notions d'astrophysiques.
- Puissance des machines mises à disposition pour l'entrainement des modèles.
- Débordement de la charge de travail car mauvaise estimation du temps qu'une ou plusieurs tâches pourraient prendre.

B.3 Gantt

Ce diagramme de Gantt visualise les tâches.

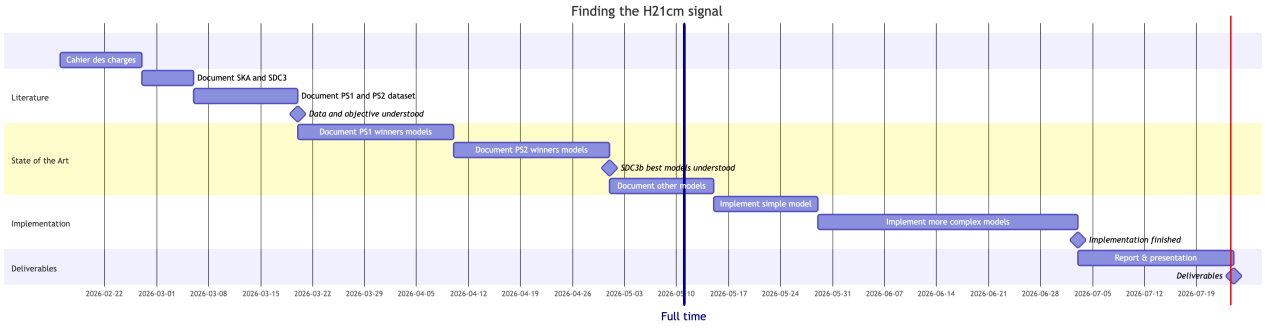
Pour la littérature, le but est de comprendre clair et net la mission du SKA, les objectifs du data challenge, et le dataset utilisé pour ce challenge (PS1 et PS2), y compris un peu de théorie d'astrophysique sur les signaux de l'hydrogène 21-cm, ceci pour concrétiser les objectifs et la relation avec le dataset.

Pour la phase de l'état de l'art, nous allons documenter, si disponible, les modèles et les papiers associés des équipes ayant eu un score 0.1 pour les deux datasets PS1 et PS2. En plus, nous allons documenter des autres types de modèles en dehors de ceux proposé par ces équipes, pour se renseigner sur d'autres possibilités d'inférence.

Pour la phase d'implémentation, l'objectif est d'itérativement développer un modèle pour inférer les spectres de puissance des signaux d'hydrogène en se basant sur les techniques découvertes lors de l'état de l'art. Les résultats de l'inférence sont donnés par l'algorithme mis à disposition par SKAO, puis nous allons les comparer avec les graphes de résultats de PS1 et PS2.

Finalement, nous allons écrire le rapport et créer les slides de la présentation pour la défense selon les directives.

Figure B.1: Gantt figure



Appendix C

Journal de travail

C.1 Week 23, 20/07 - 27/07

C.1.1 Meeting

J'ai montré les résultats et ils semblent être cohérents. Youppie!

C.1.2 Weekly

Pas de weekly

C.1.3 Work

- fini le rapport (résultats + discussion + conclusion + abstract + mise en page + bibliographie + page de titre + autres...)
- mis en annexe les valeurs de mean xHI pour tout les modèles et les scénarios + redsfhit (selon lower, mean et upper)
- Poster fait
- Code
  - commentaire
  - clean up (code et fichiers)
  - Readme
  - pousser sur Git
  - Release tag
- mis en annexe le cahier des charges, mandat et ce journal de travail. Au revoir!

C.2 Week 22, 13/07 - 19/07

C.2.1 Weekly

good advice, validation of report structuration

C.2.2 Work

- finished training all models
- meeting with expert pushed to week after because could not get results in time
- conception + implementation written, started with writing results (analysis)

C.3 Week 21, 06/07 - 12/07

C.4 Work

I done the structuration of the report, how I envision how i would write it.

Ensemble has a problem with rejection sampling. it's a bug specificed in the FAQ of sbi common with NPE (espcially when ensembling) so put the prior to be between 0 and 1 and it seems to fix it.

Problems with running:

- cuda cache clearing and deleting old arrays, density estimators, etc. in the loop of 30-run because betti could not run for longer than 2 days (but time was not the cause, it was people).
- hired hpc (h100 gpu) because people were running processes on the same gpu and crashing because OOM. Costs quite a bit and want to see if i can get renumerated for this. (in total close to 200 CHF)

For the report I wrote the state of the art.

C.5 Week 20, 29/06 - 05/07

C.6 Work

Changed to all 20 realisations instead of pick and choosing, that way it has the full variation of the observations for the same theta. It's a lot longer to train now.

Did ppc for metrics and plotted. Plot values are quite high but this is normal after doing analysis on observations (ps2d recahing up to a million in value, betti in the thousands). Plot is supposed to show for a given observation if it can map back using the parameter and to have the correct value (using KNN to find the nearest parameter). If below the diagonal line, then overestimation, underestimation if above the diagonal. PPC metrics show z-score, fractional residual (error percentage) and nearest neighbor distance. Per scenario.

Implementation is de facto done.

C.7 Week 19, 22/06 - 28/06

low, mid, high ground truth based on +- 1 sigma, could change this later thanks to the paper with real observable constraints (JWST and lyman-α)

PS2D data changed to the one given by Nicolas (it has the same indices for y as Betti)

Changed the joint loader as well

Had to change the data loader to be generic now (is cleaner).

Best model is based on the mean p-value of TARP based on 20 realisations. Also plotting 30 trials of TARP with all realizations and in green is the chosen model.

Ensemble changed to do this as well (selects 5 so 5 models are in green)

Plotted in red the ground truth for the pairplots.

C.8 Week 18, 15/06 - 21/06

C.8.1 Meeting

...

C.8.2 Weekly

... sur papier les notitions

C.8.3 Work

Problem with memmap creation of PS2D, it would duplicate the 0th index across the 150k indices. Fixed it and better PS2D performance now (it doesn't output the prior).

Started and implemented a rudimentary resnet (work in progress)

C.9 Week 17, 08/06 - 14/06

C.9.1 Weekly

- Prior mis sur 0,1 puis faire un min max de theta pour "matcher" le prior
- WST non, car chronophage

C.9.2 Work

Ensemble methods:

- take the 5 best from 30-run and create an EnsemblePosterior with it Seems to give just slightly better scores for the Betti ensemble.

Problem solving $\to$ is the data combination method a bad idea? No My reasoning:

1. Without embedding, the val loss scores of training the SBI are awful (in the positive)
2. With embedding, the scores are superior to Betti alone, implying that it's learning something extra from the PS2D data. This is also because training with our current embedding net, on PS2D alone, it's a CNN followed by a fully connected network, seems to give noise, where there is no mutual information for the SBI to learn from, and thus it just gives the prior as the posterior distribution. Surprisingly, this embedding seems to still boost the PS2D+Betti data combination.
3. With log10 of the observations of PS2D, we arrive at almost the same scores.

For the resnet idea:

1. input is PS2D with shape (3, 10, 10) where the 10x10 matrix is the k parallel and perpendicular to the line of sight, and 3 are the three redshift bins. Output is theta, where there is a value (a fraction) for each redshift, so shape (3,)
2. we train the resnet with the input to predict the based on the index since every x_obs has an index that represents a theta. And use MSE loss. Before the output we have a linear layer (should be fairly decently sized (64 neurons?)).
3. after training, we freeze the resnet's weights. also have a dropout of 0.1 to generalize.
4. as an embedding net, it will give a single output, which is what it predicts the three theta values will be, thus the summary dimension will be a vector of size 3. This way, SBI can have an estimate of the theta for a given observation which is not perfect but gives information so that it may build the posterior according to the prior, the observation, and the predicted theta.
5. With the resnet, if the training is rather fast, we can do live training during gridsearch + the 30 run. this way we don't have any data leakage.

Just need to implement it still...

C.10 Week 16, 01/06 - 07/06

C.10.1 Weekly

Sur la bonne voie

C.10.2 Work

Finished sbi pipeline, running on

- ps2d
- betti

Found TARP problem with betti, the prior was incorrectly set, but it's been rectified.
started wst implementation (from raw images).

1. iterate on raw images
2. compute wst signals
3. average down the coeffs into a 1D vector
4. save to .npy array of 1d vectors

started data combination of betti and ps2d: find pairs of indices that have matching theta. this reduces the data by a bit but by rounding to 5 decimals, we still have a dataset of length of about 137k. Created wrapper loader class to get theta, ps2d observable and betti observable (concatenation of flattened ps2d and betti obs (300+189))

Embedding is done by slicing the x and feeding the ps2d to the ps2d embedding and the same for betti, then concatenating the summary vectors of both and feeding that into a fully connected network to get the final summary vector.

C.11 Week 15, 25/05 - 31/05

C.11.1 Weekly

- R2 -¿ comment faire? Sur la bonne voie.

C.11.2 Meeting expert

posterior et histoire avec échantillons ET vrais observable (mean?)

entrainer 30 modèles -¿ tarp trop long alors skip celui là -¿ on prend celui avec le meilleur résultat (et meilleure calibration)

entrainer le embedding séparement (resnet pour CNN par exemple) régression avec un MSE -¿ réduire à trois dimensions (summary_dim=3) (z1, z2, z3) le vecteur sommaire

piocher au hasard les réalisations des bruits des Betti numbers car pour simuler le télescope

WST -¿ peut etre pas possible avec PS2D donc faut retrouver le papier référencé dans le papier de Zhao puis le partager (car sur images 2D?)

C.11.3 Work

Started Betti SBI model, embedding network is a simple fully connected network with leaky RELU and batch normalization (basically the fc part of the CNN embedding but larger).

Also ran PS2D SBI but with a log10 operation applied to it (should probably normalize it first to be positive)

Finished Betti SBI, lots of things to talk about!

Started pipeline for sbi for all data types (input in command line the data type):

1. init of things for the specified data type
2. gridsearch and cv for best hyperparameter (or input in cli the path to best params file)
3. run 30 trials of the sbi model with best params but with different train/test indices and run tarp as well (we will keep the models who can run tarp, and are in the top 5 of val_loss and tarp p-value)
4. save model with best score (judged by val loss, and tarp's atc and ks_pval)
5. full evaluation of best model

this way we have a generic way of training the best model possible.

C.12 Week 14, 18/05 - 24/05

C.12.1 Weekly

C.12.2 Work

SBI complete for PS2D

Pairplots generated for low, mid and high xHI reionization fraction to see the posterior distribution. We can surmise that...

TARP looks well calibrated (p-value=0.999...), if overconfident by a bit.

Metrics:

- General
  - R2 is close to 0 -¿ can't measure from a point since posterior DISTRIBUTION, then how to measure it?
- Per bin
  - R2 close to 0
  - RMSE and MAE seems to increase as the frequency range gets higher
- best val loss: -5.28

Embedded network tweaked:

- Batch normalization added for CNN part
- Dropout also added to reduce overfitting chances
- Extra layer in the fully connected network to make a gradual descent into the summary dimension

Reionisation history: shows too broad at the start and only constrains a bit near the end (more difficult to constrain lower frequencies?)

Uncertainty distributions for low, mid and high xHI for the three redshift bins shows lower frequency has less uncertainty.

C.13 Week 13, 11/05 - 17/05

Full time begins.

C.13.1 Weekly

No weekly, semaine autonome.

C.13.2 Work

Finished state of the art.

Restrained scope even more to focus on SBI:

- Dropped CNN to nice-to-have
- Ensemble methods are also nice-to-have

This way I can focus on training 7 models (for all combinations of data types) of SBI.

Started programming SBI on Lambda-0, made the PS2D compressor (to reduce dimensionality and to work as an embedding for the normalizing flow inside SBI). Compressor uses a small CNN network that compresses down to a specific dimension (per default 4). Instead of RELU, using GELU (G for Gaussian) as it it continuous in and around 0, which is precisely ideal since the power spectra have a lot of values around that.

C.14 Week 12, 04/05 - 10/05

C.14.1 Weekly

Short weekly.

C.14.2 Work

Continued state of the art, playing around with SBI.

C.15 Week 11, 27/04 - 03/05

C.15.1 Weekly

A en parler:

1. Changement d'horaire lors du full time?
2. Scope du projet pour ne pas trop déborder
    1. WST à heureusement une librairie python
    2. Montrer combinaisons de données et nombre de modèles à entraîner (enlever bispectrum?)
3. Les datasets doivent être mises sur le serveur lambda-0
    1. pk_128_train_250325_noise
    2. Betti numbers
4. lambda-0
    1. Refaire mot de passe (j'ai oublié).
    2. Utilisation: beaucoup de CPU et de GPU entre ˜10h-19h depuis le 15 mai (y compris le samedi mais de 11h-15h environ). Raison:
        1. Entrainement de modèle
        2. Itération rapide
        3. Données complexes et larges
    3. Authentification MLFlow pour que seulement moi puisse accéder à ça.

C.15.2 Work

Continued state of the art, should constrain the scope more since there are too many models to train and not enough time to implement all that I want to do.

C.16 Week 10, 20/04 - 26/04

C.16.1 Weekly

C.16.2 Work

Continue Zhao.

For the moment, I think I will only use the PS2D, the bispectrum and Betti numbers, even though it would be very interesting to use Zhao's WST or 3D CNN suggestion to get an all encompassing comparison between models and data combinations. -¿ restrict scope!

Just found this library that applies a WST so maybe we can apply it to the data! https://www.kymat.io/

C.17 Week 9, 13/04 - 19/04

C.17.1 Weekly

faut probablement pas faire les KANs.

C.17.2 Work

Began Zhao papre

C.18 Week 8, 06/04 - 12/04

Vacation.

C.18.1 Work

Began Cooper paper

C.19 Week 7, 30/03 - 05/04

C.19.1 Weekly

Denzel a travaillé sur les KANs et il a un étudiant en masters qui continue à travailler dessus.

C.19.2 Work

Continued paper from De Salis.

Started looking into combining multiple data sources (concatenation, etc.).

C.20 Week 6, 23/03 - 29/03

C.20.1 Weekly

Tout va bien

C.20.2 Work

Emailed Yuchen Liu from the Cantabrigians team to ask if he has a paper or link he could share with me.

Taking notes of the paper[1]. Had a meeting with the author of the paper, Nicolas Cerardi.

Started paper of supervisor on Friday.

C.21 Week 5, 16/03 - 22/03

C.21.1 Weekly

.. Laissez tel quel le bruit négatif

Log peut être marcherait moins bien

[1] https://arxiv.org/abs/2511.11568

C.21.2 Work

Trying to find white papers and/or githubs from the list of winners (score ¿ 0.1) for PS1 and PS2 datasets. A bit difficult.

Filtering SKA inference in arXiv from during/after 2025

Trouble finding white papers, found a few on arXiv, but they are not specific to the data challenge. Will email teams from the 3a challenge to see if they have anything that can help me.that can help me.

Better to start with papers that are directly related, then read the ones that are less related to the subject matter.

Links:

- https://arxiv.org/abs/2511.11568
- https://arxiv.org/abs/2310.17602
- https://scholarlypublications.universiteitleiden.nl/access/item%3A3716878/view
- https://arxiv.org/abs/2508.16329
- https://arxiv.org/abs/2508.13261 or https://arxiv.org/abs/2603.00531 (the same?)
- https://arxiv.org/abs/2407.14298
- interesting to understand h21 structure https://arxiv.org/abs/2601.13023
- dataset denoiser https://arxiv.org/abs/2507.12545

Read the first paper. Author used an SBI model with both the cylindrically averaged power spectra in addition with Betti numbers to get the best results.

Combine WST (see Zhao paper) and PS2D?

May ask for the Betti numbers dataset from the author, to have the non-Gaussian structures from the signal.

C.22 Week 4, 09/03 - 15/03

C.22.1 Weekly

A en parler: Spectrogramme est en fait une image 2D des modes de Fourier (distribution des modes représentant la puissance d'un signal) pour les lignes de visée perpendiculaires ET parallèles et groupé en k bins. ![[Pasted image 20260309094305.png|400]] Donc c'est pour ça que les inputs sont des matrices 10x10.

Les redshifts pour le y sont donc spécifique PAR plage de fréquence.

Optimization

- Opération logarithmique pour éviter le underflow
- float16 vs un nombre plus grand (float64 par exemple)
- etc.

C.22.2 Work

Using only a 30th of every power spectra file, we roughly estimate that the minimum value is about -0.15, while the max is at 7 or so. This is very rough but it gives us an idea of the range.

We could normalize the data between]0, 1] and then apply a logarithmic operation. Since a lot of values are very small, this enables us to use bigger numbers, so as to not lose precision when using float16 data types. This is a potential vector for optimization for training.

Started a data reference page https://gitlab-etu.ing.he-arc.ch/isc/2025-26/niveau-3/3285-tb-id/206/ska-sdc3b-inference/-/wikis/Data-reference

C.23 Week 3, 02/03 - 08/03

C.23.1 Weekly

redshift comme feature mais pas garantie de toujours l'avoir, ou overfitting (un peu une amélioration des performances) mais pas garder car un redshift en stack parce qu'implicitement le modèle c'est que c'est trois redshift différents (inférence par l'ordre) division par trois le sample.

Au tant garder les stacks (tester stacks + ordre implicite)

problème d'overfitting sur le data challenge mais pas sur la réalité des données de l'espace.

faire un bon score pour le challenge ou avoir un modèle plus généralisable quel que soit le redshift.

pour le nouveau dataset, on doit entrainer sur pk_res déjà fait (pk_res encore un peu bruité), mais pour le FITS cube on doit faire un peu de traitement pour que ça marche avec leur truc.

Commencer avec noise mais si le bruit est trop difficile à travailler avec, il faut faire sans noise.

train avec noise ˜160'000 samples (checker les métadonnées car ordre bizarre, répétition de pattern), train sans noise avec ˜16'000 samples

C.23.2 Work

EDA on new provided dataset (the noisy one):

- redshift/frequencies have the same min/max as the previous dataset. need to investigate what this entails, but it seems the input data has specified frequencies for each power spectra so it may not be so consequential.
- All input data (all 478350 of them) are 10x10 power spectra and there filename is formatted as such: `PS0_index_freq-min_freq-max.txt`.
- $\frac{478350}{3} = 159450$ so there are about 160 thousand samples per redshift range.
- In terms of targets, the "y_true" lists two columns and three rows:
  - `# z` which is a **specific** redshift, so not a range, but each of the three redshifts are within the three ranges (one in one). These three redshifts are *constant* for each file.
  - `mean_xHI` which is the mean value of the reionisation fraction for the specific redshift value.
    - Min values are 0 for the three values
    - Max values are north of 0.9
    - KDE graph shows the density at which point the mean reionisation fraction accumulates.
      - z_{1} fraction ...
      - z_{2} fraction ...
      - z_{3} fraction density shows that the density of the fraction is highest at about 0.9. It gradually becomes more dense from 0.0 to 0.9, then falls off very fast when nearing 1.0.

C.24 Week 2, 23/02 - 01/03

C.24.1 Weekly

Cahier de charges, il faut refaire buts pour coincider avec le Gantt

Comment évaluer les résultats: distribution, pas les moyens de reproduire Nice-to-have FITS cube résultat par SKAO

métrique intermédiaire: split données

données sont pas trop grandes, et sont créées par Michele Bianco (sur GitHub), grosses matrices.

De Salis a déposé le dataset sur le repo du projet.

C.24.2 Work

Cahier des charges has been bettered according to feedback during weekly.

Reading the dataset description.

Started an EDA on the dataset to couple with insight gained from dataset description.

C.25 Week 1, 16/02 - 22/02

Kickoff meeting the 17th. Discussed the data challenge, that we will have weekly meetings at 11h00 every Tuesday.

I will need to be pro-active and the "leader" of this project, which means I will need to stay on top of everything that happens. It's possible if the results of my work are either really promising or really terrible, that I can write an academic paper to disseminate my findings to the scientific community.

Created GitLab sub-project for the inference data challenge.

Cahier des charges and the Gantt diagram are created.

After cahier des charges and Gantt is done, I should start reading:

- SKAO (done)
- Science Data Challenge 3 (done)
- Dataset explanation for PS1 and PS2 (clean data), should take 2 or so weeks.

Git issues and milestones only after cahier des charges is validated.

Appendix D

Metrics

D.1 mean xHI

Here are metrics for the mean xHI fraction rounded to 4 decimals for all combinations of scenarios and redshift bins, for upper, mean and lower boundaries. These make up the posteriors within one standard deviation as shown in the reionization history plots in the main report document. In order these are the single-model PS2D, Betti and PS2D + Betti, and then their ensembles.

Table D.1: PS2D mean x_{HI} fraction and credible boundaries

Scenario and z bin	Lower	Mean	Upper x_{HI}
Low z_1	0.5382	0.5614	0.5833
Low z_2	0.3333	0.3558	0.3795
Low z_3	0.0835	0.1062	0.1283
Mid z_1	0.7454	0.7581	0.7717
Mid z_2	0.6160	0.6307	0.6463
Mid z_3	0.4751	0.4945	0.5140
High z_1	0.9052	0.9185	0.9322
High z_2	0.8253	0.8456	0.8662
High z_3	0.7006	0.7317	0.7621

Table D.2: Betti mean x_{HI} fraction and credible boundaries

Scenario and z bin	Lower	Mean	Upper x_{HI}
Low z_1	0.5441	0.5546	0.5651
Low z_2	0.3272	0.3368	0.3466
Low z_3	0.0973	0.1003	0.1033
Mid z_1	0.7292	0.7354	0.7420
Mid z_2	0.5985	0.6076	0.6168
Mid z_3	0.4581	0.4675	0.4773
High z_1	0.9046	0.9133	0.9226
High z_2	0.8275	0.8395	0.8519
High z_3	0.6955	0.7100	0.7250

Table D.3: PS2D + Betti mean $x_{\rm HI}$ fraction and credible boundaries

Scenario and z bin	Lower	Mean	Upper $x_{\rm HI}$
Low z_1	0.5374	0.5501	0.5627
Low z_2	0.3211	0.3315	0.3420
Low z_3	0.0972	0.1017	0.1062
Mid z_1	0.7423	0.7502	0.7579
Mid z_2	0.6094	0.6184	0.6274
Mid z_3	0.4736	0.4844	0.4953
High z_1	0.8959	0.9059	0.9155
High z_2	0.8190	0.8326	0.8462
High z_3	0.6985	0.7147	0.7304

Table D.4: Ensemble PS2D mean $x_{\rm HI}$ fraction and credible boundaries

Scenario and z bin	Lower	Mean	Upper $x_{\rm HI}$
Low z_1	0.5313	0.5568	0.5819
Low z_2	0.3247	0.3502	0.3775
Low z_3	0.0818	0.1062	0.1313
Mid z_1	0.7361	0.7521	0.7680
Mid z_2	0.6022	0.6219	0.6415
Mid z_3	0.4589	0.4837	0.5085
High z_1	0.9043	0.9192	0.9350
High z_2	0.8234	0.8476	0.8732
High z_3	0.6993	0.7387	0.7805

Table D.5: Ensemble Betti mean $x_{\rm HI}$ fraction and credible boundaries

Scenario and z bin	Lower	Mean	Upper $x_{\rm HI}$
Low z_1	0.5427	0.5550	0.5667
Low z_2	0.3246	0.3370	0.3489
Low z_3	0.0960	0.1001	0.1042
Mid z_1	0.7231	0.7338	0.7446
Mid z_2	0.5903	0.6024	0.6144
Mid z_3	0.4527	0.4654	0.4782
High z_1	0.8934	0.9060	0.9187
High z_2	0.8137	0.8301	0.8468
High z_3	0.6900	0.7077	0.7264

Table D.6: Ensemble PS2D + Betti mean $x_{\rm HI}$ fraction and credible boundaries

Scenario and z bin	Lower	Mean	Upper $x_{\rm HI}$
Low z_1	0.5430	0.5578	0.5721
Low z_2	0.3256	0.3374	0.3496
Low z_3	0.0947	0.0999	0.1050
Mid z_1	0.7267	0.7372	0.7477
Mid z_2	0.5934	0.6051	0.6169
Mid z_3	0.4565	0.4685	0.4808
High z_1	0.8905	0.9023	0.9141
High z_2	0.8089	0.8256	0.8422
High z_3	0.6885	0.7069	0.7248

D.2 30-trial run TARPS

Here are the trial runs' TARP calibrations for those not shown in the main body.

Figure D.1: Ensemble PS2D 30 run trial TARP calibrations

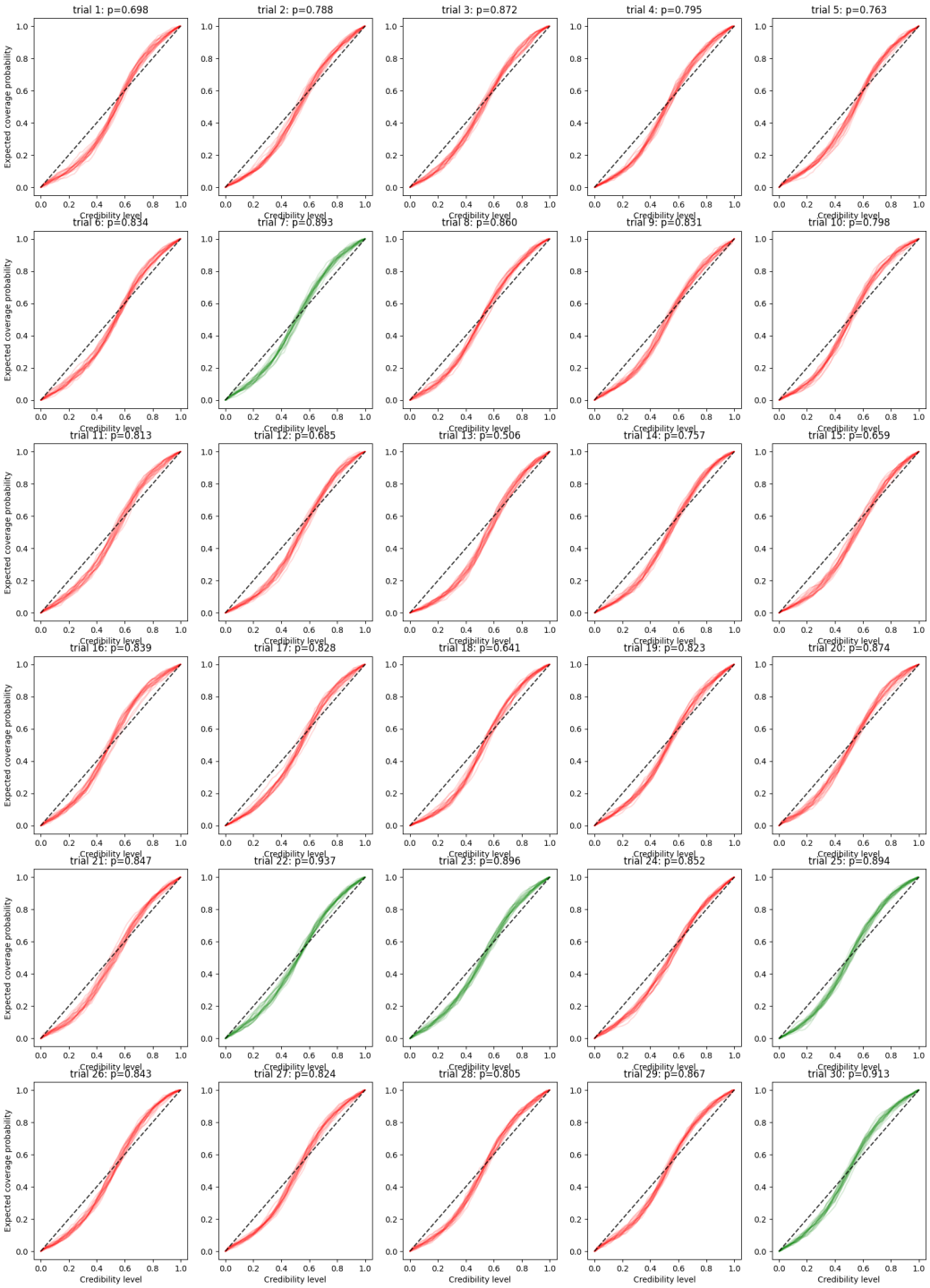


Figure D.2: Ensemble Betti 30 run trial TARP calibrations

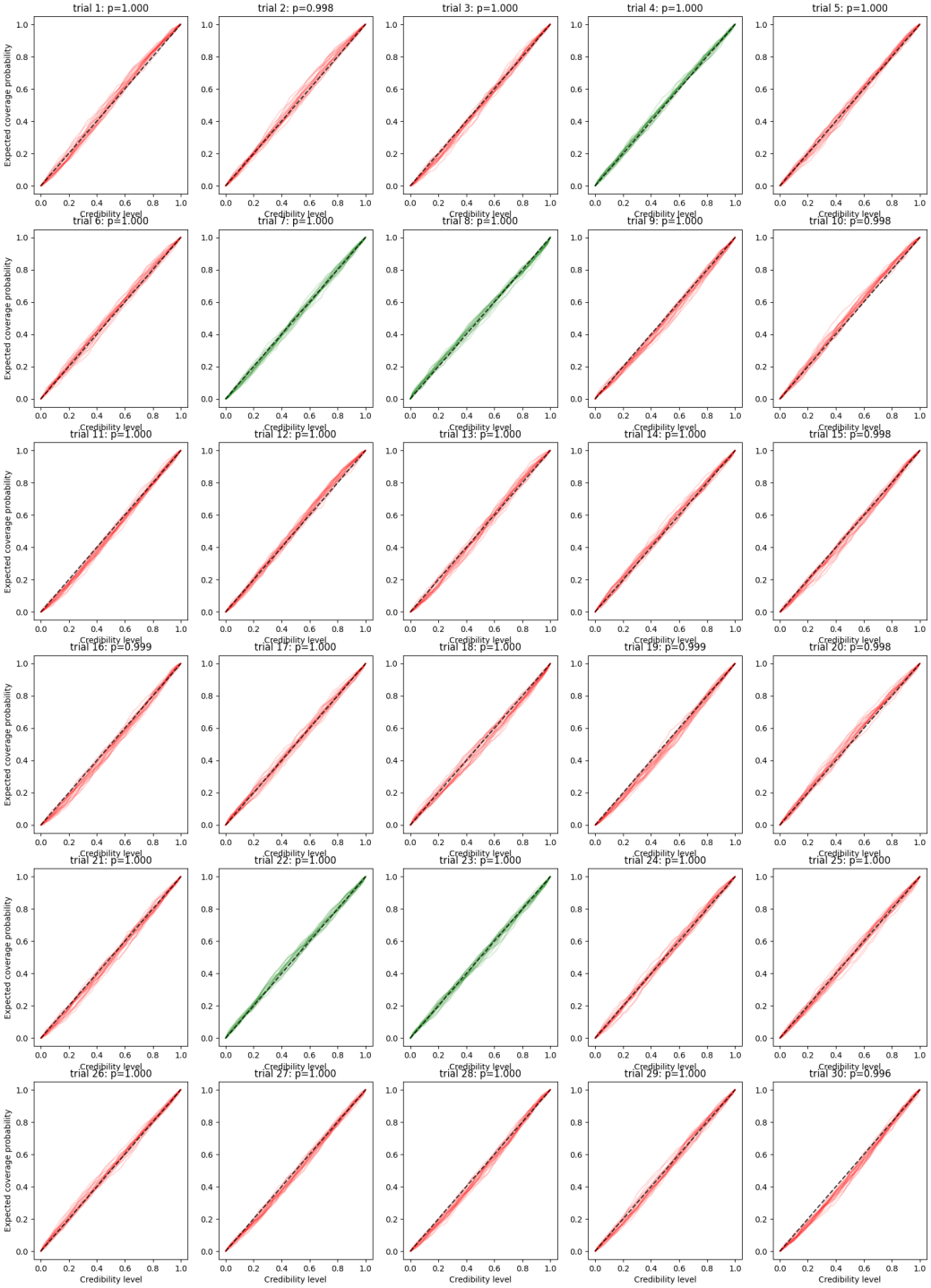


Figure D.3: Ensemble PS2D + Betti 30 run trial TARP calibrations

